W209 Final Presentation

NBA Shot Chart Visualization

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Visualizing NBA Player Stats: Background

Data

- Official NBA API (GitHub)

Users / Stakeholders

- Teams
- Players
- Scouts
- Fans

Tasks / Goals of Visualization

- Provide insight into player shooting efficiency across the court
- Broad player shooting trends over time

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Data Assembly & Cleaning

```
#import needed package from API
import nba_api
from nba_api.stats.static import players
from nba_api.stats.endpoints import playergamelog
from nba_api.stats.library.parameters import SeasonAll
from nba_api.stats.endpoints import playbyplay
import pandas as pd
import json
import requests
import datetime
import numpy as np
pd.set_option('display.max_columns',None)
```

Process:

- Obtained seasonal data from NBA Shot Chart & Player Game Log API endpoints

- Requested data from API in batches due to large quantities, stored in multiple Pandas dataframes, before combining

- **Cleaned or added fields such as:**
  - Team names (home/away, player team vs opponent)
  - Game date formats
  - Seasons
  - Player names (in player data)
  - Time remaining per game

- **End result:**
  - Two large datasets exported to csv for upload to Tableau
  - NBA_shot_data
    - Details each shot instance, including player and team info, shot type, location, and make/miss info going back to 1996-97 season
    - 5M+ rows; ~1 GB
  - NBA_player_data
    - Details game logs by player id for each game appearance since 1996-97. Team matchup, and individual game stats (offensive and defensive)
    - 600k+ rows; ~100 mb

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Usability Study: Findings and Resolutions

- Initial Figma shot chart model

- Multiple shot charts or an ability to toggle between shot charts for various representations like total shot attempts, total made shots, and percent efficiency to strengthen the shot chart page

- Ability to view shooting efficiency based on opponent or timing within the season (playoffs vs regular season)

- Add a legend to the shot chart and use an easily differentiable color scheme

- Streamline the organization of the basic statistics dashboard

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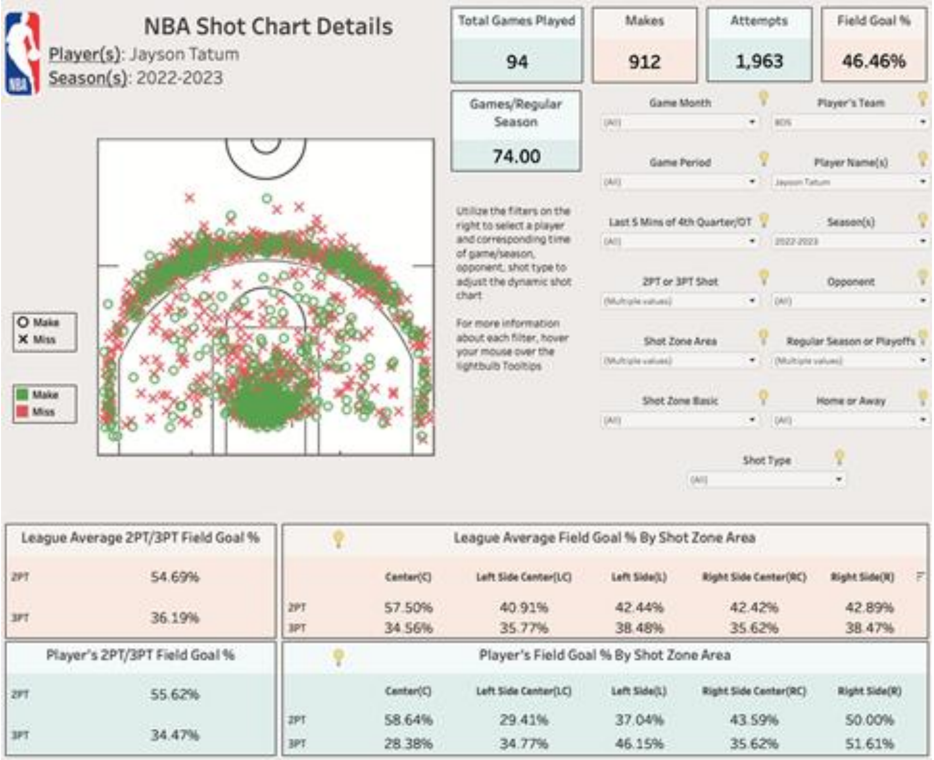
Final Player Shot Chart

- **Final Product**
  - [Link to Interactive Shot Chart]()



Player Stat Dashboard (Link)

- User presented view of various aggregate statistics encompassing all areas of game across seasons
  - Shooting (attempts, makes, %)
  - Other offense (assists, offensive rebounds)
  - Defense and misc. (blocks, steals, turnovers)
- ‘All’ is initial view – user prompted to select a statistic which will then populate the dashboard

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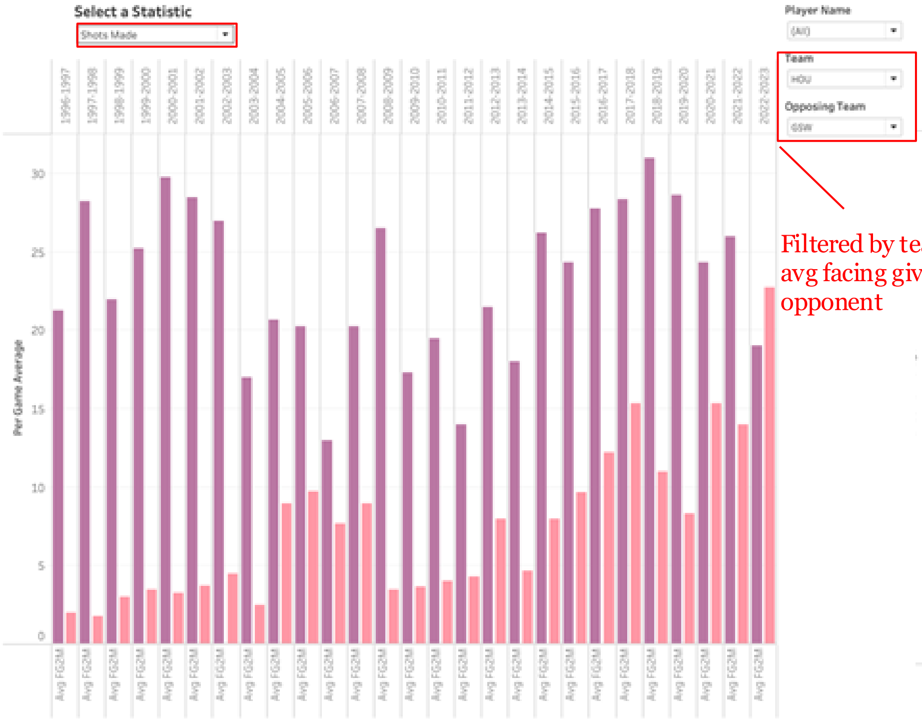
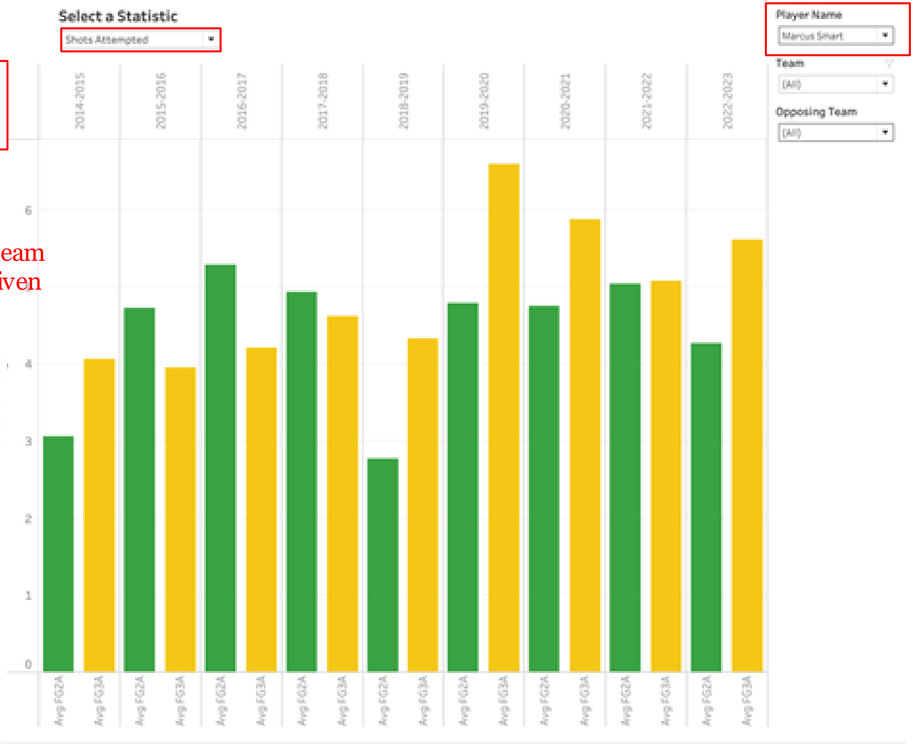
Player Stats Cont.



Filtered by team avg facing given opponent

Filtered by player



Tools

Version Control
- Team GitHub Page

Data Preprocessing
- Jupyter Notebook / Google Colab

Dashboards/Visuals
- Figma
- Tableau

Website
- Flask, HTML/CSS

Data Source & Team

- **Sources**
  - Shot Chart API: https://github.com/swar/nba_api/blob/master/docs/nba_api/stats/endpoints/shotchartdetail.md
  - Game Log API: https://github.com/swar/nba_api/blob/master/docs/nba_api/stats/endpoints/playergamelog.md
  - Player IDs API: https://github.com/swar/nba_api/blob/master/docs/nba_api/stats/static/players.md
- **Team**
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  - Daniel Myers: dm2714@berkeley.edu
  - David Sherman: davidsherman@berkeley.edu
  - Savinay Chandrupatla: savinay@berkeley.edu
- **Demo Video**
- **Complete Presentation Video**

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